Bio203 objective notes for mid term

Subject: Bio203 (Notes)

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From topics 1 to 4

1) A technique widely used in Molecular Biology and Biotechnology is called_____?
   a) PCR B) Primer c) DNA polymerase d) Taq polymerase
2) During PCR, a heatstable DNA polymerase, such as Taq polymerase - an enzyme derived from the_____?
   A) Template b) primer
   c) bacterium Thermus aquaticus d) DNA polymerase.
3) PCR can be performed to amplify _____?
   A) DNA B) RNA C) A&B D) none of these
4) First regular cycling event and consists of heating the reaction to 93°C - 98°C for 20-45 seconds in____?
   A) Denaturation B) Annealing
   C) Extension D) ALL of these
5) ANNEALING step has The reaction temperature ?
   a)93°C - 96°C b) 93°C - 98°C c) 40-65°C d) 70-80°C.
6) what step depends on the type of DNA polymerase used in PCR?
   A) Denaturation B) Annealing C) Extension
   D) ALL of these
7) Taq polymerase has its optimum activity temperature at_____ ?
   a)93°C - 96°C b) 93°C - 98°C c) 40-65°C d) 70-80°C.
   (its optimum activity temperature.Commonly 72°C is used)
8) In EXTENSION step the DNA polymerase synthesizes a new DNA strand complementary to the DNA template strand by adding ?

A) RNA B) DNA C) dNTP's D) DNA polymerase

9) FINAL ELONGATION TEMPERATURE is_____?

a) 93°C - 96°C b) 93°C - 98°C c) 40-65°C d) 70-74C.

10) Which of the following is a DNA POLYMERASES?

A) Taq polymerase b) Pfu polymerase

c) Vent polymerase D) ALL OF these

11) Primers are single stranded DNA fragments of the size_____?

A) 18–30 bp long b) 20–30 bp long

c) 18–50 bp long d) 20–50 bp long

12) Distance between the primers binding sites will determine the –_____product?

A) size of DNA B) SIZE of RNA

C) size of PCR D) SIZE of primer

13) Primer PRODUCT SIZE?

A) 100–700 bp B)(100–800 bp C) (100–900 bp D)(100–1000 bp

14) IN primer the Quantity of G/C content is_____?

A) 30–80%; B) 40–80%; C)20–80%; D) 20–90%;

15) Primers determine the specificity ?

A) PCR reaction B) DNA C) RNA D) DNA polymerase

16) How mamy steps are held in PCR reaction?

A) 2. B) 3 c) 4. D) 5.

17) Define PCR?

ANS: A technique widely used in Molecular Biology and Biotechnology. PCR can be performed to amplify DNA. It can be extensively modified to perform a wide array of genetic manipulations

18) WRITE THE FORMULA FOR CALCULATING MELTING TEMPERATURE OF PRIMERS ?

ANS:Tm = 4(G+C) + 2(A+T)

19) NAME THE INGREDIENTS OF PCR?

Microfuge tube, Thermal cycler, DNA template, Primers, Buffer, MgCl2, Distilled water, Deoxynucleotide triphosphates and DNA polymerase

20) HOW MANY STEP of PCR reaction must write their names?
Ans:THREE STEPS
Denaturation , Annealing & Extension

From Topic 4 to 8

21)The most commonly used polymerase in pcr is?

a) Taq polymerase b) Pfu polymerase c)Ventpolymeras **d)** all of these

23) what type of polymerase used in Reverse transcriptase for activity?

a) Taq polymerase **b) Pfu polymerase**

c)Vent polymerase **d) Tth polymerase**

24)how many cycle are required for standard PCR?

a)30 -40 cycleb) 30 -50 cycle c)30-60 cycle d) 30-70 cycle

25) The amplification program consists of a series of ?

a)30 -40 cycle b) 30 -50 cycle c)20–50 cycle d) 20 -60 cycle

26)The function of Annealing temperature is?

a) Controls specificity of hybridization

b)Optimized for individual polymerases

c) Reduces double stranded molecules to single stranded

d) all of these

27)what is the function of Denaturation temperature?

a) Controls specificity of hybridization

b)Optimized for individual polymerases

c)Reduces doublestranded molecules to single stranded

d) all of these

28) Primer dimers may occur due to_____?

a) reduction of primers
b) Controls specificity
c) Optimization of primers
D) hybridization of primers

29) ____ _is the Controls for contamination and contains all reagents except DNA template?

a) BLANK REACTION
b) hybridization of primers
c) Annealing temperature
d) NEGATIVE CONTROL

30) _____is the Controls for specificity of the amplification reaction and contains all reagents and a DNA template lacking the target sequence?

a) BLANK REACTION
b) hybridization of primers
c) Annealing temperature
d) NEGATIVE CONTROL

31) RAPD PCR (AP-PCR) are used for _____ ?

a) Quantitative trait
b) qualitative trait
c) above all
d) none of these

32)____ is the type Of PCR in which two type of primer are present?

a)NestedPCR
b) Multiplex PCR
c) touchdown PCR
d) Sequence-specific PCR

33)What type of PCR in which RNA convert in DNA and than DNA becomes cDNA?

A) Reversetranscriptase PCR
b) Multiplex PCR
c) touchdown PCR
d) Sequence-specific PCR

34) what are TYPES OF PCR?

ANS; Nested PCR, Multiplex PC, Multiplex PC , Touchdown PCR, Sequence-specific PCR, Reversetranscriptase PCR, Long-range PCR , Whole-genome amplification, RAPD PCR (AP-PCR, Quantitative real-time PCR

35) WHAT IS BLANK REACTION ?

ANS:Controls for contamination and contains all reagents except DNA template.

36) WHAT IS NEGATIVE CONTROL ?

ANS: Controls for specificity of the amplification reaction contains all reagents and a DNA template lacking the target sequence

From TOPIC 9 TO 12

37) What type Of PCR in which two type of primer are present?

a) Nested PCR
b) Multiplex PCR
c) Touchdown PCR
d) Sequence-specific PCR

38) In Nested PCR the SECOND PCR PRODUCT IS SPECIFIC to increas the specificity ?

a) DNA amplification
b) RNA amplification
b) C) ALL OF these
d) none of these

39)what are ADVANTAGE OF NESTED PCR ?

Ans: Very low probability of nonspecific amplification

39) Define MULTIPLEX PCR?

ANS:Multiplex PCR is a variant of PCR which enables simultaneous amplification of many targets of interest in one reaction by using more than one pair of primers.

40)Annealing temperatures for each of the primer sets_____to work correctly within a single reaction?

A) must be HIGH B) must be LOW

C) must be optimized D) NONE OF THESE

41) Amplicon sizes should be different enough to form distinct bands when visualized by_____?

ANS) Gel electrophoresis

42) The primer design for all primers pairs has to be optimized so that all primer pairs can work at the _____during PCR. ?

A) same annealing temperature B) D/F annealing temperature
C) NONE OF THESE d) both of these

43) IN MULTIPLEX PCR The use of multiple, unique primer sets within a single PCR reaction is to produce?

ANS: amplicons of varying sizes specific to different DNA sequences

44) If amplicon sizes overlap, the different amplicons may be differentiated and visualised using_____?

a)template b) pcr c) **cycle** **d) primers**

45)The primer design for all primers pairs has to be optimized so that all primer pairs can work at the same annealing temperature during?

a)template b) pcr c) **cycle** **d) primers**

46) Based on the process of reverse transcription, which reverse transcribes_____?

A)RNA into DNA B)DNA into RNA C)BOTH of these d) none of these

47) Synthesis of cDNA using oligo dT primers ?

A) (38°C) one hour. B) (37°C) one hour. C) (40°C) one hour. D) (45°C) one hour.

48)ENZYME REVERSE TRANSCRIPTASE IN The PCR is preceded by a reaction using reverse transcriptase to convert ?

A)RNA into DNA B)DNA into RNA C) RNA to cDNA. D) DNA to cDNA.

49)REVERSE TRANSCRIPTASE PCR To identify the sequence of an ?

A) RNA transcript. B) DNA transcript. C) tRNA transcript. D) mRNA transcript.

50) Define REAL TIME PCR?

Ans: Real Time PCR is a technique in which fluoroprobes bind to specific target regions of amplicons to produce fluorescence during PCR

51)Quantitative real-time PCR is often confusingly known as ?

A) RT-PCR (Real Time PCR B) RQPCR

C) BOTH OF THESE D) none of these

52)QRT-PCR methods use containing DNA probes,?

a)fluorescent dyes b) SybrGreen, c) fluorophore d) all of these

53 to measure the amount of amplified product in real time we use ?

A) Primers B)Buffer C) MgCl2 D)TaqMan

From Topic 13 to 14

54) This is a technique that reduces nonspecific amplification during the initial set up stages of the PCr ?

A) Nested PCR b) Multiplex PCR c) -PCR (Real Time PCR d)HOT START PCR

55) IN HOT START PCR The technique may be performed manually by heating the reaction components upto the melting temperature____

A) 95°C B) 96°C C) 98°C D) 99°C

56) _____is used to amplify one strand of the original DNA more than the other?

A) Asymmetric PCRb) Multiplex PCR c) -PCR (Real Time PCR
D)) Nested PCR

57) _____used in some types of sequencing and hybridization probing where having only one of the two complementary strands is ideal ?

A) Asymmetric PCRb) Multiplex PCR c) -PCR (Real Time PCR
D)) Nested PCR

58)What are uses of REAL TIME PCR?

Ans:Real Time PCR is used to measure the quantity of a PCR product. The fluorescence, measured in Real Time, is detected in a PCR cycler with an inbuilt filter flurometer. It is the method of choice to quantitatively measure starting amounts of DNA, cDNA or RNA.

From Topic 15 to 18

59) Extended or longer than standard PCR, meaning over ____ kilo bases (frequently over 10 kb). ?

a) 7 b) 6 c) 5 d) 15

60)Special mixtures of proficient polymerases along with accurate polymerases such as are often mixed together?

a) Taq polymerase b) Pfu polymerase

c) Vent polymerase D) ALL OF these

61) LONG PCR to clone large genes not possible with ?

A)RT PCR B) NESTED PCR

C) conventional PCR. D) NONE OF THESE

62)Wite the Application - LONG PCR ?

Ans; Application - to clone large genes not possible with conventional PCR.

63)Allele-specific PCR used for identifying of_____ ?

Ans: SNPs

64) ALLELE SPECIFIC PCR Uses primers whose_____ends encompass the SNP?

A) 3 B) 5 C) BOTH of these d) none of these

65)ALLELE SPECIFIC PCR in This diagnostic or cloning technique is used to identify or utilize ?

Ans: singlenucleotide polymorphisms (SNPs) Allele

66)define COLONY PCR?

Ans:The screening of bacterial or yeast clones for correct ligation or plasmid products

67)write the METHODOLOGY of the COLONY PCR?

Ans:The screening of bacterial or yeast clones for correct ligation or plasmid products. Pick a bacterial colony with an autoclaved toothpick, swirl it into 25 µl of TE autoclaved dH2O in an microfuge tube. Heat the mix in a boiling water bath (90-1000C) for 2 minutes Spin sample for 2 minutes high speed in centrifuge.Transfer 20 µl of the supernatant into a new microfuge tube. Take 1-2 µl of the supernatant as template in a 25 µl PCR standard PCR reaction.

68)In Situ PCR (ISH) is a polymerase chain reaction ~~that~~ actually takes place _____on a slide.?

a) inside the cell b) outside the cell

c) both of these d) none of these

69)IN SITU PCR Limited to detection of non-genomic material such as____?

A)DNA B) DNA C)t RNA D) RNA.

From Topic 19 to 26

70) is used to identify patterns of DNA methylation at CpG islands in genomic DNA ?

a) Inter sequence pcr b) suicide pcr

c) Methylation-specific PCR d) all of these

71)Two)amplifications are then carried out on the bisulfite-treated DNA: One primer set anneals to DNA with (corresponding to methylated cytosine), and the other set anneals to DNA with (corresponding to unmethylated cytosine).

a) cytosines and uracil b) uracil and cytosine
c) guanine and adenine d) adenine and guanine

72) Where used Methylation Specific PCR?

Ans) MSP used in quantitative PCR provides information about methylation state of a given CpG island.

73) A method for DNA fingerprinting that uses primers selected from segments repeated throughout a genome to produce a unique fingerprint of amplified product lengths.

a)Intersquencepcr b) suicide pcr

c) Methylation-specific PCR d) INTER SEQUENCE SPECIFIC PCR

74)uses of InterSequence Specific PCR?

Ans) PCR used to produce a unique fingerprints of amplified product lengths.

75) Write Ligation-mediated PCR METHODOLOGY?

Ans)Uses small DNA oligonucleotide 'linkers' (or adaptors) that are first ligated to fragments of the target DNA. PCR primers that anneal to the linker sequences are then used to amplify the target fragments.

76)What are uses of Ligation-mediated PCR?

Ans)DNA sequencing Genome walking DNA footprinting

77) Primers can be designed to be 'degenerate' - able to _____from a large number oftarget locations?

a) initiate replication
b) middle replication
c)final replication
d) all of these

78)IN PCR Whole genomes can be amplified by ?

A)a)Intersquencepcr

b) suicide pcr

c) Methylation-specific PCR

D) Whole Genome Amplification PCR

79) Polymerase Chain Reaction is also called technique of molecular biology used in ?

a) Molecular Biology
b)Biotechnology
c) both of these
d) none of these

80) first time DNA polymerase was purified by ?

Ans) Thermus aquaticus

81)polymerase chain has its components name ?

Ans: DNA polymerase

82)what is step in which double standard DNA convert in single standard ?

a) Denaturation b) Annealing c) Extension d)all of these

83)in which step primer bind with the single stand ?

a) Denaturation b) Annealing

c) Extension d)all of these

84)extention step is also called ?

a) Denaturation b) Annealing

c) Elongation d)extention

85) DNA is used in pcr as ?

a) Primer b) template c) gene d) all of these

86) is used to identify patterns of DNA methylation at CpG islands in genomic DNA ?

a) Intersquencepcr b) suicide pcr

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92)what are uses of Ligation-mediated PCR?

Ans)DNA sequencing Genome walking DNA footprinting

93) Primers can be designed to be 'degenerate' - able to from a large number of target locations?

a) Initiate replication b) stop replication

c) Finalize replication d) all of these

94)IN PCR Whole genomes can be amplified by ?

A)a)Intersquencepcr b) suicide pcr

c) Methylation-specific PCR D) Whole Genome Amplification PCR

95)Real Time PCR is used to measure the of a PCR product ?

a) quality b) quantity c) specificity d) all of these

96)The fluorescence, measured in Real Time, is detected in a with an inbuilt filter flurometer.

a) thermal cycle b) Annealing cycle C) PCR cycler D) none

(THANK You Remember US In your PraYerS)

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